Pre-gravity6plane: the Pin(4,4) fermion field dirac16complex, its 6-planes, gravity, Kohn-Sham pair creation, Einstein-Rosen bridges and deflating degrees of freedom

This repository extends the Mathematica notebook Pre-gravity6plane.nb (P. L. Nash, (C) 2022, GNU General Public License) with a second-quantized fermion field, dirac16complex, and the calculations requested for it. Every equation and result stated below is computed with exact arithmetic (or a convergent numerical method where stated) in the notebooks 6planes.nb (Sections 1 to 4) and deflation.nb (Section 5). This notebook is distributed as free software under the GNU General Public License.They are distributed in the hope that they will be useful,but WITHOUT ANY WARRANTY;without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.See the GNU GPL for more details.

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1. Setting

Coordinates $x_0, \ldots, x_7$ (x_0 hidden space, $x_{1,2,3}$ space, x_4 time, $x_{5,6,7}$ superluminal time), metric $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$. The eight real 16×16 Dirac matrices are those of Pre-gravity6plane.nb,

$$T^A = (T16^A)[A] = \begin{pmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{pmatrix}, \qquad T^A T^B + T^B T^A = 2\eta^{AB} I_{16}, \qquad S^{AB} = \tfrac{1}{4}[T^A, T^B],$$

with entries in $\{-1, 0, 1\}$; the S^{AB} generate Spin(4,4) and, with the reflections T^C, Pin(4,4); $\Gamma = T^0 \cdots T^7 = \mathrm{diag}(-I_8, I_8)$, $P_{L,R} = \frac{1}{2}(I_{16} \mp \Gamma)$, $\sigma_{16} = T^0 T^1 T^2 T^3$.

2. The field and its Lagrangian

dirac16complex $\psi = (\psi_0, \ldots, \psi_{15})$ has complex anticommuting (Grassmann) components and is a second-quantized fermion field; $\bar{\psi} = \psi^\dagger \sigma_{16}$.

$$\mathcal{L} = \tfrac{1}{2}\left(\bar{\psi} T^A \partial_A \psi - (\partial_A \bar{\psi}) T^A \psi\right) - m\bar{\psi}\psi - \frac{\lambda}{2}(\bar{\psi}\psi)^2, \qquad T^A \partial_A \psi = (m + \lambda \bar{\psi}\psi)\psi.$$

$\mathcal{L}$ is exactly real; the quadratic (mass) and quartic (self-coupling) terms are non-zero because the components are complex. Quantization with time x_4 gives $\{\psi_a, \psi_b^\dagger\} = K_{ab}\delta$, $K = -i\sigma_{16} T^4$, $K^2 = I$, with an (8,8) indefinite (Krein) metric, and an exact split into two mirror sectors of opposite energy: $\mathcal{L} = \mathcal{L}_D[\psi_+] - \mathcal{L}_D[\psi_-] - \frac{\lambda}{2}(S_+ - S_-)^2$ for fields independent of x_5, x_6, x_7.

3. Mirror of Pre-gravity6plane.nb and the 6-planes

Restricted to configurations that depend only on x_0 and x_4, the analogue of the notebook's $\Psi_{16}(x_0, x_4)$ ($m = -2M$), the 16 field equations split into the four blocks $\{0,5,8,13\}, \{1,4,9,12\}, \{2,7,10,15\}, \{3,6,11,14\}$, as in the old notebook; its exponential solutions $e^{pz+qt}u$ with $(6p+q)(6p-q) = m^2/H^2$, $z = 6Hx_0$, $t = Hx_4$, are exactly the Maple solutions of the old notebook. With the triality frames and vector forms $V_1 = \eta(\psi_1^T \sigma {F_A}^a)$, $V_2 = {F_a}^A \psi_2$, a Universe defines the 6-plane

$$M^6 = \{Z \in \mathbb{R}^8 : n_1 \cdot Z = 0,\ n_2 \cdot Z = 0\}, \qquad n_k = V_k[\mathrm{Re}\,\psi],$$

with $\binom{8}{2} = 28$ parameterizations; a pair of universes of energies $\pm E$ creates the 6-plane $\{Z : \mathrm{Re}V_2 \cdot Z = \mathrm{Im}V_2 \cdot Z = 0\}$ by itself.

4. Gravity, Kohn-Sham DFT, pair creation, Einstein-Rosen bridges

Three gravitational fields (diagonal vielbeins): primordial $\mathrm{diag}(\tan 6Hx_0,\ e^{-Hx_4}\sin^{-1/6} 6Hx_0\ (\times 3),\ 1,\ e^{Hx_4}\sin^{-1/6} 6Hx_0\ (\times 3))$ (the vielbein of Pre-gravity6plane.nb with $a_4 = Hx_4$); modern

diag$(f^{-1/2},1,1,1,f^{1/2},1,1,1)$ with $f=1-r_s/x_0$; future the same with $f=1-r_s/x_0-H_\Lambda^2x_0^2$. In each, the Euler-Lagrange equations derived from the Grassmann Lagrangian are $\gamma^\mu D_\mu\psi=(m+\lambda\bar\psi\psi)\psi$, e.g. (primordial)

$$\cot(6Hx_0)T^0\partial_0\psi+e^{Hx_4}\sin^{\frac16}(6Hx_0)\sum_{i=1}^{3}T^i\partial_i\psi+T^4\partial_4\psi$$
$$+e^{-Hx_4}\sin^{\frac16}(6Hx_0)\sum_{j=5}^{7}T^j\partial_j\psi-3H\cot^2(6Hx_0)T^0\psi=(m+\lambda\bar\psi\psi)\psi.$$

The quartic term is treated in a Kohn-Sham (DFT) approximation, $M=m+\lambda\rho_s$, with the Kohn-Sham Hamiltonian $h_{KS}=\alpha p+BM\beta$ ($\alpha=-T^4T^0$, $\beta=-iT^4$) in tortoise coordinates, ground and first-excited (ΔSCF) states, and the finite-temperature thermodynamic effective potential $\Omega(T)$. Proven in each field: $\mathcal{L}_m[\Gamma\psi^*]=\mathcal{L}_{-m}[\psi]$ and $\mathcal{L}_m[T^1\psi(R_1x)]=\mathcal{L}_{-m}[\psi]$ (mass-sign pairing); $T^0h_{KS}T^0=-h_{KS}$ and $h_{KS}^*=-h_{KS}$ (every level ϵ has a partner $-\epsilon$); in the zero-energy vacuum the creation of a + universe pair together with a − universe pair changes the energy by exactly 0, while single quanta and single-sign creation are forbidden by the conserved charges and energy: universes of masses $\{+m,-m\}$ are created in pairs. The 1935 Einstein-Rosen bridges are verified; the modern field has one and the future field two exact bridges (two congruent sheets, $g=0$ on the bridge, finite curvature); the primordial field's degenerate hypersurface $x_0=0$ is a curvature singularity ($R\approx-\frac{31}{216H^2x_0^4}$); across a bridge the vacuum pairs universes of energy $+\epsilon$ and $-\epsilon$ with weight $1/(e^{\epsilon/T_H}+1)$, $T_H=\kappa/2\pi$.

5. Deflating degrees of freedom

In deflation.nb the superluminal times x_5,x_6,x_7 of the primordial field are replaced by deflating degrees of freedom ($z=6Hx_0$):

$$ds^2=\tan^2z\,dx_0^2+e^{-2Hx_4}\sin^{-1/3}z\,(dx_1^2+dx_2^2+dx_3^2)-dx_4^2-g_{55}\,dx_5^2-g_{66}\,dx_6^2-g_{77}\,dx_7^2,$$
$$g_{jj}=b_{jj}e^{-ax_4},\quad a>0,\ b_{jj}>0.$$

Proper sizes along x_5,x_6,x_7 shrink as $\sqrt{b_{jj}}\,e^{-ax_4/2}$. The Ricci scalar, the Einstein tensor and the Kretschmann scalar depend on x_0 only, so deflation creates no curvature singularity at finite x_4. The pure deflation field $dx_0^2+dx_1^2+dx_2^2+dx_3^2-dx_4^2-\sum_j g_{jj}dx_j^2$ is flat $\mathbb{R}^4$ times a hyperbolic 4-space of radius $2/a$ whose four directions are all timelike,

$$-dx_4^2-\sum_{j=5}^{7}g_{jj}\,dx_j^2=-\frac{4}{a^2}\,\frac{dw^2+dy_5^2+dy_6^2+dy_7^2}{w^2},\qquad w=\frac{2}{a}\,e^{ax_4/2},\quad y_j=\sqrt{b_{jj}}\,x_j,$$

with $R=3a^2$ and Kretschmann scalar $3a^4/2$. World lines with momentum $p_5=-g_{55}\,dx_5/d\ell$ (ℓ an affine parameter) turn back in time at $x_4^*=\frac1a\ln(b_{55}/p_5^2)$. The field equation of dirac16complex, derived exactly from the Grassmann Lagrangian, is

$$\cot z\,T^0\partial_0\psi+e^{Hx_4}\sin^{\frac16}z\sum_{i=1}^{3}T^i\partial_i\psi+T^4\partial_4\psi+e^{ax_4/2}\sum_{j=5}^{7}\frac{T^j\partial_j\psi}{\sqrt{b_{jj}}}$$
$$-\frac32H\cot^2z\,T^0\psi-\frac32\left(H+\frac{a}{2}\right)T^4\psi=(m+\lambda\bar\psi\psi)\psi.$$

Its universes $e^{i(k_0x_0+k_1x_1+k_2x_2+k_3x_3+q_5x_5+q_6x_6+q_7x_7-\omega x_4)}u$ (frozen coefficients) obey

$$\omega^2=m^2+\cot^2z\,k_0^2+e^{2Hx_4}\sin^{1/3}z\,(k_1^2+k_2^2+k_3^2)-e^{ax_4}\sum_{j=5}^{7}\frac{q_j^2}{b_{jj}},$$

so momentum along a deflating direction turns oscillation into exponential growth, and the critical deflation rate is $a_c=2H$. Every mode equation $\chi'=d\chi/dx_4=M\chi$ with $M=T^4(i\sum_{A\neq4}c_AT^A-m)$ and real c_A satisfies $M^\dagger K+KM=0$. So the Krein charge $\chi^\dagger K\chi=P_+-P_-$ is conserved, where $P_\pm=\chi^\dagger\frac12(I_{16}\pm K)\chi$ are the weights of the +mass and -mass sectors, while only the deflating directions change the norm P_++P_-. Hence P_+ and P_- grow by equal amounts: deflation creates {+mass, -mass} universes in pairs. In the exact numerical example the -mass fraction $P_-/(P_++P_-)$ grows from 1.2×10^{-31} to 0.49999995.

6. Assumptions and interpretations made in this work

1. The modern and future gravitational fields (not specified in the request) are defined as above; the primordial field is the notebook's vielbein with $a_4(Hx_4) = Hx_4$.
2. "Universe" means a mode function (c-number solution) of dirac16complex, as the notebook calls Ψ_{16} the wave function of a Universe; its "mass" is its energy for translations of x_4 (or the mass parameter m for the mass-sign theorem).
3. The theorems are exact statements about this model; they are not a proof about the physical universe.
4. In Section 5 the deflating components enter the line element as $-g_{jj}dx_j^2$ and all other components are those of the primordial field, as the user specified; the dispersion relation there uses frozen (adiabatic) coefficients.

7. Reproducing everything

```
cd code
wolframscript -file verify_dirac_matrices.wls ../Pre-gravity6plane.nb
wolframscript -file build_nb2.wls
wolframscript -file build_deflation.wls
python3 gen_dm.py
python3 md2tex.py <file>.md <file>.tex ; pdflatex <file>.tex
```

8. Repository inventory

file	content
Pre-gravity6plane.nb	original notebook (P. L. Nash, GPL)
6planes.nb	new notebook, evaluated, 34 input cells, no messages
deflation.nb	new notebook on the deflating degrees of freedom, evaluated, 17 input cells, no messages
README.md/.tex/.pdf	this overview
dirac matrices.md/.tex/.pdf	proofs and display of the eight real 16x16 Dirac matrices
dirac16complex_6planes.md/.tex/.pdf	the field, Lagrangian, quantization, mirror of the old notebook, 6-planes
gravity_DFT_pair_creation.md/.tex/.pdf	three gravitational fields, Euler-Lagrange equations, Kohn-Sham DFT, pair theorems
einstein_rosen_bridges.md/.tex/.pdf	Einstein-Rosen bridges
deflation.md/.tex/.pdf	deflating degrees of freedom: geometry, curvature, geodesics, field equation, universes, pair creation
*_provenance.md/.tex/.pdf	sources and computations behind each document
*_symbols.md/.tex/.pdf	definitions of all symbols of each document
code/	core.wl, verify_dirac_matrices.wls (+ log), dm_data.json, gen_dm.py, md2tex.py, 6planes_src.txt, build_nb2.wls (+ log), deflation_head.txt, deflation_tail.txt, deflation_src.txt, build_deflation.wls (+ log)

Pre-Universe

[Whenever possible, we try to follow ROBERT L. BRYANT , 'SUBMANIFOLDS AND SPECIAL STRUCTURES ON THE OCTONIANS', J. DIFFERENTIAL GEOMETRY 17 (1982) 185-232; any ERRORS ARE DEFINITELY due to ME]

Let $\mathbb{R}^{4,4}$ denote a real, eight-dimensional, pseudo-Euclidean vector space endowed with the $SO(4,4;\mathbb{R})$ invariant quadratic form

$x0^2 + x1^2 + x2^2 + x3^2 - x4^2 - x5^2 - x6^2 - x7^2$,

which defines the standard pseudo-Euclidean norm on $\mathbb{R}^{4,4}$ and associated non-degenerate indefinite inner product.

Let $\overline{SO(4,4;\mathbb{R})}$ denote the universal cover of $SO(4,4;\mathbb{R})$.

Spinor notation:

"left-handed" and "right-handed" spin(8) spinors have $\overline{SO(4,4;\mathbb{R})}$ counterparts that are denoted as ψ_1 and ψ_2 in this repo.

In this draft the phrase 'Pre-Universe'

describes a conjectured 'Quantum construction' of a Pre-Big-Bang ***spacetime***,

built using

the split octonion algebra $O_{4,4}$

and

the (pseudo-) Clifford algebra $C_{4,4}$ (with multiplicative identity id) generated by a set of 8 elements $e_A^{4,4},\ A = 1-1, ...8-1$ that anticommute with each other and have square equal to $\pm$ id.

Explicitly, the generators $e_A^{4,4},\ A = 0, 1, ..., 7$ satisfy

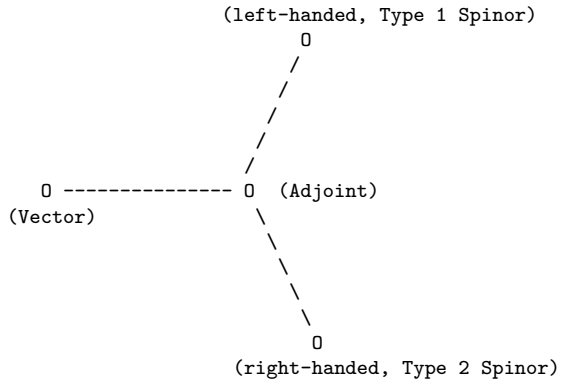
$$e_A^{4,4} e_B^{4,4} + e_B^{4,4} e_A^{4,4} = id\ g_{AB},\ A, B = 0, 1, ..., 7$$

where

$g_{AB} = diag(1, 1, 1, 1, -1, -1, -1, -1)$.

The (scaled) skew-symmetrized products of the $e_A^{4,4}$, formed from an irreducible representation of the $e_A^{4,4}$, are the infinitesimal generators of a $16 = 2^{8/2}$-dimensional , reducible, basic spinor representation of $\overline{SO(4,4;\mathbb{R})}$.

Dynkin Diagram for D_4 [SO(8) ; Spin(8)]



Node Legend:

- **Central Node:** Adjoint representation
- **Left Node (180°):** Vector representation
- **Top-Right Node (60°):** Type 1 Spinor (left-handed spinor) representation
- **Bottom-Right Node (-60°):** Type 2 Spinor (right-handed spinor) representation

We represent spinors in this 16-component basic spinor representation as

$\psi \in \mathbb{R}^{16} = \{\psi_1, \psi_2\}$,

which are realized according to

$$\psi \in \mathbb{R}^{16} = \{\psi_1 \in \mathbb{R}^8 \oplus \psi_2 \in \mathbb{R}^8\}$$

$$\psi_1 = \psi_1^a, a = 0, 1, ..., 7$$

$$\psi_2 = \psi_2^{(a)}, (a) \to a = 0, 1, ..., 7$$

[see split octonions, $O_{4,4}$ = Os, below];

A Lagrangian

for ψ is defined,

Euler-Lagrange equaions solved,

vector forms of ψ_1 and ψ_2 calculated (please see 'split octonions' , below),

two 7-planes, respectively dual to the preceding two vectors, are computed,

the 6-plane of their intersection identified;

todo:

associate 6-plane with 3 generations of Einstein-Rosen 2-planes.

This

study of a possible 'Quantum construction' of a Pre-Big-Bang ***spacetime*** is also carved out using the split octonion inner product algebra

$O_{4,4}$ = Os

and Cartan's 'triality' .

Recall:

several properties of the split octonion inner product algebra Os are::

$1 \in Os$;

non-degenerate indefinite inner product

$$<,>: Os \times Os \mapsto \mathbb{R}$$

and a multiplication $Os \times Os \mapsto Os$ with unit $1 \in Os$ such that

$\forall x, y \in Os$, < x y, x y > = <x, x> <y, y>.

<1, 1> = 1;

For convenience's sake we will identify $\mathbb{R}$ with the 1-dimensional subalgebra of Os generated by $1 \in Os$.

We define the orthogonal compliment of 1 to be $Im\ Os \in Os$. It is a proper subspace and we have $Os = \mathbb{R} \oplus Im\ Os$.

Given $x \in Os$, we define the conjugate x^* according to

$x \mapsto x^* = 2 < x, 1 > -x$;

We denote <x, 1> by Re x

and (x - Re x) by Im x.

Clearly $x \in Im\ Os$ if and only if $x^* = -x$ or x = Im x or Re x = 0.

Let

$e_A,\ A = 0, 1, ..., 7$, comprise a Minkowski vector basis of Os.

We are using the convention that

$$e_0 = 1$$

A 'Minkowski' vector realization of $x \in Os$ is

$x = x^A e_A,\ A = 0, 1, ..., 7$ where

$x^A \in \mathbb{R}$.

(equivalent [by Cartan's triality] to each of the two independent spinor bases; explicit realizations of equivalence are constructed in this repo's files),

and verify

$< e_A, e_B >= g_{A,B} = diag(1,1,1,1,-1,-1,-1,-1)$,

Os is endowed with a non-associative multiplication $Os \times Os \mapsto Os$, which may be defined according to

$e_A \ e_B = e_C \ m^C_{A \ B}$.

In this repo we construct the multiplication constants $m^C_{A \ B}$ from

[1] our irrep of the Clifford algebra $C_{4,4}$ generators ; and

[2] a type 1 spinor u,

verifying < u, u > = 1.

SPLIT OCTONION MULTIPLICATION TABLE $e_A \ e_B$

A/B	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	$-e_3$	e_2	$-e_5$	e_4	e_7	$-e_6$
e_2	e_2	e_3	$-e_0$	$-e_1$	$-e_6$	$-e_7$	e_4	e_5
e_3	e_3	$-e_2$	e_1	$-e_0$	$-e_7$	e_6	$-e_5$	e_4
e_4	e_4	e_5	e_6	e_7	e_0	e_1	e_2	e_3
e_5	e_5	$-e_4$	e_7	$-e_6$	$-e_1$	e_0	$-e_3$	e_2
e_6	e_6	$-e_7$	$-e_4$	e_5	$-e_2$	e_3	e_0	$-e_1$
e_7	e_7	e_6	$-e_5$	$-e_4$	$-e_3$	$-e_2$	e_1	e_0

Let

$e^{1/2}_a, \ a = 0, 1, ..., 7$ comprise a type 1 spinor basis of Os.

A type 1 spinor $\Psi_1 \in Os$ may be realized according to

$\Psi_1 = \psi^a_1 \ e^{1/2}_a, a = 0, 1, ..., 7$, where

$\psi^a_1 \in \mathbb{R}$.

Similarly, a type 2 spinor $\Psi_2 \in Os$ may be realized according to

$\Psi_2 = \psi^{(a)}_2 \ e^{1/2}_{(a)}, \ (a) = (0), (1), ..., (7)$, where

$\psi^{(a)}_2 \in \mathbb{R}$, and

and the $e^{1/2}_{(a)}, \ (a) = (0), (1), ..., (7)$ comprise a type 2 spinor basis of Os.

We restrict Spin(4,4) ($= \overline{SO(4,4)}$) transformations to the subgroups preserving both $\mathbb{R}$ and Im Os, i.e., restrict to Spin(3,4)

in which case Ψ_1 and Ψ_2 transform the same way under 'allowed Lorentz' transformations.

For brevity, understanding this restriction, $(a) = (0), (1), ..., (7) \ \rightarrow a = 0, 1, ..., 7$.

We are beginning to study 3-Generations of Einstein-Rosen 2-Plane-Bridges

BibTeX: @article{Einstein Rosen 1935,author={Einstein,A.and Rosen,N.},title={The Particle Problem in the General Theory of Relativity},journal={Phys.Rev.},volume={48},issue={1},pages={73--77},year={1935},publisher={American Physical Society},doi={10.1103/PhysRev.48.73}},

for the case of Einstein-Rosen 2-Plane-Bridges $\in \ Os$

The origin of this calculation has its roots in "The Particle Problem in the General Theory of Relativity" and "A Remarkable Representation of the 3+2 de Sitter Group" by P.A.M.Dirac, J.Math.Phys.4,901[Dash]909 (1963).

The term "reduced Brauer[Dash]Weyl generators", refers to a specific set of irreducible generators for certain complex Clifford algebras, particularly in the context of mathematical physics and the study of Dirac spinors. These generators were introduced by E.A. Lord (E. A. Lord. "The Dirac spinor in six dimensions". Mathematical Proceedings of the Cambridge Philosophical Society, vol. 64, pp. 765[Dash]778, 1968) and are described as "reduced" because they form an irreducible representation of the Clifford algebra Subscript[[ScriptCapitalC], 2 n] from Subscript[[ScriptCapitalC], 2 (n-1)] .

Here, we employ Lord's idea of reduced Brauer[Dash]Weyl generators to extend Dirac's 4x4 gamma matrices into this notebook's real tau8 8x8 and tau16 16x16 matrices, which are employed to write down the 'Dirac Equation for the Pre-Universe' [remember, this is before the Big Bang, and before the particles of the standard model exist] formulated in terms of a 16-component Overscript[O(4,4), _] spinor [CapitalPsi]16 ([CapitalPsi]16 AKA 'WAVE FUNCTION OF the UN-UNIVERSE,' which could possibly be named something more pompous). This equation

is used to investigate EINSTEIN AND ROSEN's idea: "If one tries to interpret the regular solution (5a) in the space of r, 0, ø, t, one arrives at the following conclusion. The four-dimensional space is described mathematically by two congruent parts or "sheets," corresponding to u>0 and u <0, which are joined by a hyperplane r= 2m or u=0 in which g vanishes .2 We call such a connection between the two sheets a "bridge.""

PLEASE IGNORE, FOR NOW, THE FOLLOWING:

Notation: $M_{4,4}$ = Minkowski spacetime, signature = {4,4};

$O_{4,4}$ = Os =split octonion algebra;

with vector and spinor bases: o[Element]Subscript[[DoubleStruckCapitalO], 4,4]; o =Subscript[eA, j] oˆj=Subscript[ea[Psi]1, j] o[Psi]1ˆj=Subscript[ea[Psi]2, j] (o[Psi]2ˆj) [Cartan's triality] =Subscript[[Psi]1, j] eaˆj=Subscript[ea, j] ([Psi]2ˆj) [restrict to Spin(3,4)];

<Subscript[ea, a],Subscript[ea, b]> = Subscript[[Sigma], ab] <Subscript[eA, A],Subscript[eA, B]> = Subscript[[Eta], AB] {X,EA}[Element]Subscript[M, 4,4][CircleTimes]Subscript[[DoubleStruckCapitalO], 4,4]